

U-Net, LW-Net and Siamese Network in Retinal Biometric Recognition

Martin Nachtmann, Miloš Oravec, Veronika Kurilová

Faculty of Electrical Engineering and Information Technology,

Slovak University of Technology in Bratislava, Ilkovičova 3, 812 19 Bratislava, Slovakia

milos.oravec@stuba.sk

Abstract—This paper presents a comprehensive study on retinal biometrics, highlighting the effectiveness of retinal vessel patterns as reliable biometric identifiers due to their unique and stable characteristics. We provide an extensive review of retinal anatomy, publicly available datasets, and current state-of-the-art segmentation and verification methods. The study emphasizes the importance of accurate vessel segmentation, evaluating U-Net and LW-Net neural networks, and investigates identity verification performance using Siamese networks based on a ResNet50 backbone with contrastive loss. The final configuration achieved an Area Under the Curve (AUC) of 0.9792, an F1-score of 90.8%, and a False Positive Rate (FPR) of 7.1%.

Keywords—Retina; Biometrics; Vessel Segmentation; U-Net; LW-Net; Siamese Networks

I. INTRODUCTION

Biometric recognition aims to identify or verify individuals based on their unique physiological or behavioral traits. Among physiological modalities, the retina stands out due to the complex and highly individual vascular structure located at the back of the eye. Unlike other traits, such as fingerprints or facial features, the retina is well protected within the body, making it difficult to forge or alter, and providing great long-term stability.

Although retinal biometrics has been known since the early 20th century, its practical deployment was historically limited by expensive hardware and low user convenience. Recent advances in imaging technology and deep learning have revived interest in this field, making automated retinal recognition more feasible [1].

This paper aims to design essential components of retinal biometrics into a practical evaluation pipeline. It covers anatomical insights, dataset characteristics, vessel segmentation using deep neural networks (U-Net and LW-Net), and a Siamese verification model based on contrastive learning. The study focuses on evaluating segmentation quality and biometric verification performance, highlighting both successful and suboptimal experimental outcomes.

The rest of the paper is structured as follows: Section II details the methodology. Section III describes the datasets. Section IV presents segmentation results, Section V covers identity verification, and Section VI concludes the work.

II. METHODOLOGY

A. System Overview

The proposed biometric system is structured as a pipeline consisting of four stages: image acquisition, preprocessing, vessel segmentation, and identity verification. Input fundus images are first enhanced using several preprocessing techniques. Subsequently, vessels are segmented using deep learning models. The resulting binary vessel maps are then passed to a Siamese neural network to verify whether two samples belong to the same individual.

B. Retinal Anatomy and Vessel Relevance

The retina is a multi-layered neural tissue located at the back of the eye. It contains key anatomical landmarks relevant for biometric identification, including the optic disc, macula, fovea, arteries, and veins. These structures form a complex vascular network that is unique to each individual, even among identical twins [2].

Figure 1 highlights the main regions involved in biometric processing. The vascular pattern originating from the optic disc is particularly important for segmentation and identity verification.

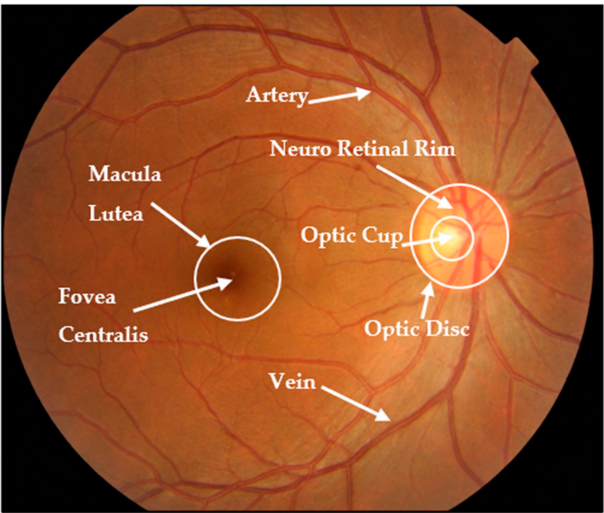


Figure 1. Annotated retinal image showing key anatomical landmarks [3].

C. Image Preprocessing

Preprocessing is a crucial step to ensure consistency and enhance the visibility of retinal vessels across varying image qualities.

In our pipeline, each fundus image is first converted to grayscale and normalized. To emphasize vascular structures, contrast enhancement techniques are applied, followed by optional smoothing. Most importantly, fixed-size patches are extracted from the full images to reduce computational complexity and increase data diversity during training. This patch-based approach also improves generalization by allowing localized learning on vessel-rich regions.

These preprocessing steps ensure that the segmentation models receive input with consistent contrast, scale, and structural detail.

D. Segmentation and Classification Pipeline

The core processing pipeline is composed of:

- 1) **Segmentation using U-Net and LW-Net**, producing binary vessel masks from preprocessed images.
- 2) **Thresholding and binarization**, fine-tuned to preserve vascular structure.
- 3) **Siamese network-based verification** using a ResNet50 encoder and contrastive loss to measure similarity.

The segmentation outputs from the segmentation process are passed to a Siamese verification model, allowing a direct comparison of segmentation quality and its impact on identity verification performance.

III. DATASETS

Several publicly available retinal fundus image datasets were utilized for training and evaluating vessel segmentation models, as well as for biometric verification. Each dataset varies in resolution, annotation quality, and the presence of ground truth vessel masks.

A. Training and Segmentation Evaluation

DRIVE [4] and **STARE** [5] are widely used for benchmarking vessel segmentation models. Both contain high-resolution color fundus images with manually annotated vessel masks, and are suitable for supervised learning and validation of segmentation quality.

CHASE_DB1 [6] was primarily used for training both U-Net and LW-Net in this work. It contains 28 retinal images with corresponding ground truth masks annotated by experts. The dataset features high vessel density and clear anatomical structure, making it ideal for detailed vessel segmentation.

The **AFIO** dataset [7] (Annotated Fundus Images for Ophthalmology) was used for segmentation testing. It contains 100 color retinal images alongside manually annotated binary vessel masks. Unlike other datasets, AFIO is particularly suitable for this task due to high-resolution images and annotations.

B. Other Datasets

Several other datasets, including **RIDB** [8], **DRiDB** [9], as well as **ARIA**, **VARIA**, **SiMES**, **Messidor**, and **IDRiD**, were also considered during the evaluation process. However, many of these datasets are primarily intended for clinical tasks such as image registration or diabetic retinopathy grading. They often lack consistent and high-quality ground truth vessel masks, clear subject-level data separation, or were unavailable due to missing public links or unresponsive authors. As such, they were deemed unsuitable for biometric segmentation and verification in this work.

IV. VESSEL SEGMENTATION

A. U-Net Architecture

U-Net is a convolutional neural network originally designed for biomedical image segmentation [10]. Its symmetric encoder-decoder architecture with skip connections enables precise localization while preserving spatial context, which is essential for segmenting complex vessel structures in retinal fundus images.

In this work, U-Net was trained using 48x48 pixel patches extracted from the mentioned datasets. The training process was carried out for 30 epochs with a batch size of 128 using the Adam optimizer. Binary cross-entropy loss was minimized to learn a clear vessel-background distinction. A public implementation of U-Net was used as the base, some modifications were made.

The model was evaluated on several retinal image databases. Table I summarizes segmentation performance using common metrics, including sensitivity (Sn), specificity (Sp), accuracy (Acc), F1-score, and AUC (* segmentation on the AFIO database was performed by a model trained on the CHASE database).

TABLE I. U-NET SEGMENTATION RESULTS ACROSS DATASETS

Dataset	F1	Sn	Sp	Acc	AUC
DRIVE	0.8169	0.7728	0.9826	0.9559	0.9794
STARE	0.8219	0.7739	0.9867	0.9638	0.9846
CHASE	0.7891	0.7482	0.9810	0.9521	0.9696
AFIO*	0.7534	0.7033	0.9389	0.9272	0.9283

Figure 2 shows a qualitative example from the CHASE_DB1 dataset with the original image, ground truth, and U-Net prediction.

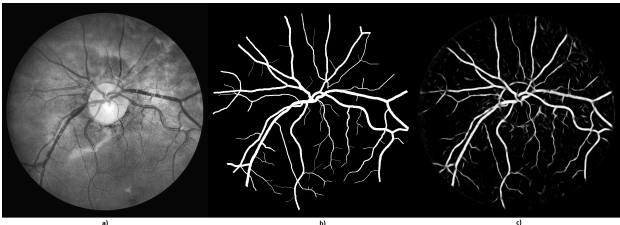


Figure 2. CHASE_DB1 segmentation: (a) Original, (b) Ground Truth, (c) U-Net Prediction.

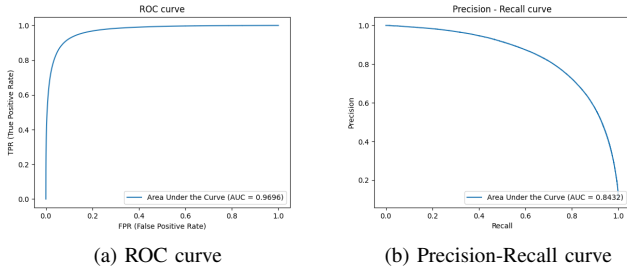


Figure 3. U-Net segmentation on CHASE_DB1: (a) ROC curve, (b) Precision-Recall curve.

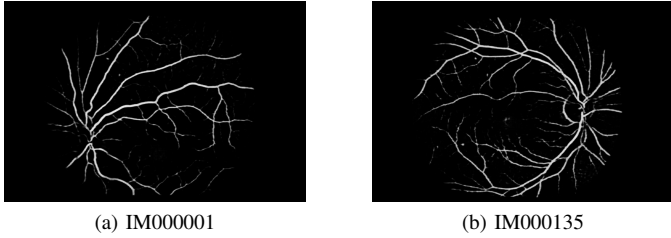


Figure 4. Example U-Net segmentation on unseen AFIO dataset.

The segmentation performance was also evaluated using ROC and Precision-Recall curves. Results on CHASE_DB1 are presented in Figure 3.

To test generalization, the model trained on CHASE_DB1 was applied to the AFIO dataset, which features different resolution and image properties. Despite no fine-tuning, the results showed reasonable segmentation quality, as shown in Figure 4.

Although the performance on AFIO was slightly lower compared to training-domain datasets, it demonstrated that the model could transfer to unseen domains with minimal degradation. This supports the use of U-Net for biometric pipelines where generalization is essential.

B. LW-Net Architecture

LW-Net (Lightweight Network) is a segmentation architecture designed to balance accuracy and computational efficiency. It is composed of two cascaded U-Net-like sub-networks: the first network performs coarse segmentation while the second refines the output, enabling detailed vessel reconstruction with sharper boundaries and less noise [11].

Unlike classic U-Net, which directly outputs binarized masks or requires threshold tuning, LW-Net produces a soft grayscale probability map that retains finer tonal gradients. These outputs can be thresholded post hoc to obtain binary masks suitable for feature extraction and biometric verification tasks.

Figure 5 shows example output from the DRIVE dataset. Although quantitative metrics were similar to U-Net, LW-Net consistently delivered cleaner, more visually accurate masks—especially in capturing fine capillaries and reducing border artifacts.

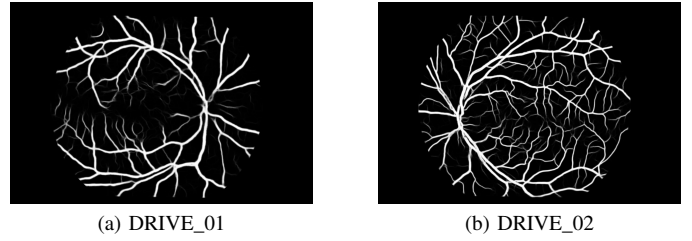


Figure 5. Example LW-Net output on DRIVE images. Binarized masks retain detailed vascular structure with less noise.

C. Summary

LW-Net offers an effective alternative to U-Net for retinal vessel segmentation. While performance metrics such as accuracy and AUC are comparable, the visual quality of LW-Net outputs is often superior, producing sharper vessel edges and capturing fine structural details. This makes it well-suited for applications where tasks, like biometric matching, depend on precise vascular geometry.

V. SIAMESE NETWORK FOR IDENTITY VERIFICATION

The final step of the biometric pipeline involves verifying whether two segmented retinal vessel maps originate from the same individual. To accomplish this, we implemented a Siamese neural network leveraging a frozen ResNet50 backbone, trained to distinguish identity based on vascular structure similarity.

A. Architecture and Training

The Siamese model consists of two identical branches sharing weights. Each branch takes a segmented vessel map (224×224) as input and passes it through a frozen ResNet50 backbone pretrained on ImageNet. The feature map is then processed by a small projection head consisting of a Global-AveragePooling layer, two fully connected layers ($256 \rightarrow 128$), ReLU activation, dropout (rate=0.3), and L2-normalization to produce a compact 128-dimensional embedding. The final output is the Euclidean distance between embeddings, representing the dissimilarity score.

Contrastive loss was used as the training objective, minimizing distances between “same” pairs and maximizing distances between “different” pairs with a margin of 1.0. The training set was augmented with random rotations, flips, brightness and contrast shifts to improve generalization. Training used the AdamW (variant of Adam with decoupled weight decay) optimizer with early stopping and learning rate scheduling.

To simulate multiple samples per person in DRIVE, we applied zoom and shift augmentations. To ensure person-disjoint evaluation, we used a 70/15/15 split on DRIVE subjects for training, validation, and testing respectively.

B. Thresholding and Binarization

Initial experiments using grayscale vessel segmentations as input yielded suboptimal results, suggesting that noise and grayscale ambiguities negatively impact model learning. We

VI. CONCLUSION

This paper presented a complete pipeline for biometric person recognition based on retinal vessel patterns. We evaluated the role of accurate vessel segmentation using U-Net and LW-Net architectures, demonstrating their impact on the identity verification task.

Experimental results confirmed that binary vessel maps serve as effective input for a Siamese neural network. The final system achieved an AUC of 0.979 and an F1-score of 90.8%, showing that vascular geometry can reliably differentiate between individuals, even across different datasets.

We also explored the effects of binarization thresholds on classification performance, finding that suppressing fine vessel noise improved discriminative accuracy.

Future work may include incorporating optic disc segmentation to remove false positives in the disc area, which may degrade the verification performance. Additionally, testing the proposed Siamese verification framework on a larger and more diverse dataset is crucial to confirm its generalizability beyond the limitations of DRIVE.

ACKNOWLEDGMENT

This paper was supported by the APVV grant no. APVV-22-606 AI4EYE and VEGA grant no. 1/0202/23 AIDabiomeDIA, both of the Slovak grant agencies.

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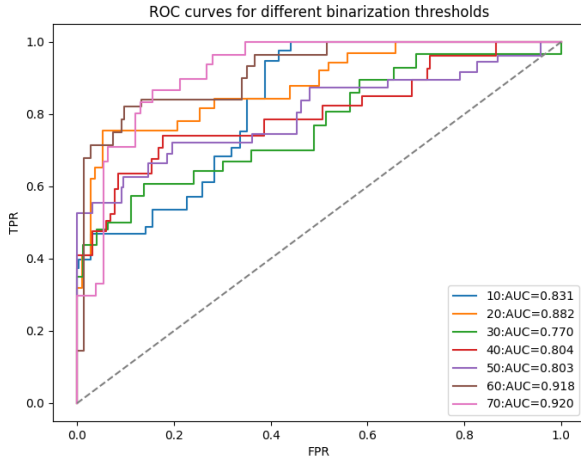


Figure 6. ROC curves for different binarization thresholds. Threshold 70 yielded the highest AUC.

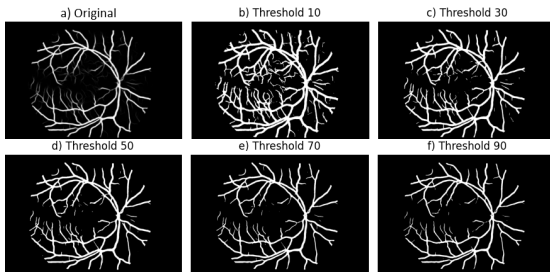


Figure 7. Effect of binarization thresholds on vessel segmentation. (a) Original grayscale; (b-f) Thresholds 10 to 90. Higher thresholds retain dominant vessels while suppressing noise.

hypothesized that binary vessel maps would reduce noise and emphasize topological structure.

To test this, we binarized vessel masks using static thresholds and evaluated their impact on verification accuracy. The experiment spanned multiple thresholds from 10 to 90, with ROC curves plotted for each (Figure 6). A threshold of 70 yielded the best performance, with AUC = 0.920. Visual inspection (Figure 7) confirmed that higher thresholds preserved the main vascular structure while suppressing spurious fine vessels and noise, improving inter-class separability.

C. Final Results and Evaluation

The final model used a binarization threshold of 70/255 and achieved excellent performance on unseen subjects. ROC curves and confusion matrix analysis confirmed strong separation between identity classes.

- **AUC:** 0.979
- **F1-score:** 90.8%
- **Precision:** 92.5%
- **Recall:** 89.3%
- **FPR at TPR = 90%:** 7.1%

These results validate the use of binarized vessel maps and contrastive learning as a powerful framework for retinal identity verification.